

```
-- events
-- A MySQL event is a scheduled task that runs automatically at a specific time
-- or repeatedly at defined intervals.
-- instead of running a query manually, MySQL can execute it automatically based
-- on time.
```

```
-- event_scheduler
show variables like 'event_scheduler';
```

```
-- to turn it on temporarily
set event_scheduler = 1;
set event_scheduler = on;
```

```
-- to turn it on permanently
set global event_scheduler = 1;
set global event_scheduler = on;
```

```
-- syntax
create event event_name
on schedule time_schedule
do
sql_statement;
```

```
-- example
create event delete_old_records
on schedule every 1 day
do
delete from logs where created_at < now() - interval 30 day;
```

```
-- Types of events:-
-- 1) one time event :- runs only once at a specific time
-- 2) Recurring events :- runs repeatedly at a fixed interval.
```

```
-- one time events:-
-- syntax
```

```
create table test_table (emp_id int,
name varchar(225), age int);
```

```
insert into test_table values(1,"abc",63),
(2,"dfs",34),
(3,"Sdfsd",45),
(4,"fsd",63),
(5,"hfg",56),
(6,"der",67);
```

```

create event retire_employees
on schedule at '2026-03-16 14:49:00'
do
delete from test_table where age > 60;

-- how to see your events
show events;

-- 2) Recurring Events:-
-- syntax
create event event_name
on schedule every interval
do
sql statment;

create table log_table(log_id int,
created_at timestamp default current_timestamp);

create event insert_logs
on schedule every 30 second
do
insert into log_table(log_id) values(rand() * 99);

show events;

select * from log_table;

-- how to disable an event
alter event insert_logs disable;

show events;

-- how to drop an event
drop event insert_logs;

-- recurring event with starts
create event insert_logs
on schedule every 30 second
starts '2026-03-16 15:02:00'
do
insert into log_table(log_id) values(rand() * 99);

select * from log_table;

-- recurring event with start and end

```

```

create event insert_logs
on schedule every 30 second
do
insert into log_table(log_id) values(rand() * 99);

select * from log_table;

show events;

-- altering an event

alter event insert_logs
on schedule every 1 minute;

-- Hospital_db
create database hospital_db;
use hospital_db;

CREATE TABLE patients (
    patient_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    email VARCHAR(150) UNIQUE,
    admitted_at DATETIME,
    status VARCHAR(20)
);
CREATE TABLE medicines (
    medicine_id INT PRIMARY KEY AUTO_INCREMENT,
    name VARCHAR(100),
    stock INT,
    expiry_date DATE
);
CREATE TABLE appointments (
    appointment_id INT PRIMARY KEY AUTO_INCREMENT,
    patient_id INT,
    doctor VARCHAR(100),
    scheduled_at DATETIME,
    status VARCHAR(20),

    FOREIGN KEY (patient_id) REFERENCES patients(patient_id)
);
CREATE TABLE event_log (
    log_id INT PRIMARY KEY AUTO_INCREMENT,
    event_name VARCHAR(100),
    ran_at DATETIME,
    message TEXT
);

```

```

INSERT INTO patients (name, email, admitted_at, status) VALUES
('Rahul Sharma','rahul@gmail.com','2026-03-01 09:00:00','admitted'),
('Priya Verma','priya@gmail.com','2026-03-02 11:30:00','discharged'),
('Amit Singh','amit@gmail.com','2026-03-03 14:00:00','admitted'),
('Neha Kapoor','neha@gmail.com','2026-03-03 10:15:00','admitted'),
('Vikas Mehta','vikas@gmail.com','2026-03-04 16:20:00','discharged'),
('Anjali Gupta','anjali@gmail.com','2026-03-05 08:40:00','admitted'),
('Rohit Jain','rohit@gmail.com','2026-03-06 13:10:00','admitted'),
('Sneha Reddy','sneha@gmail.com','2026-03-06 17:25:00','discharged'),
('Karan Malhotra','karan@gmail.com','2026-03-07 09:45:00','admitted'),
('Pooja Sharma','pooja@gmail.com','2026-03-08 12:30:00','admitted'),
('Arjun Nair','arjun@gmail.com','2026-03-08 15:00:00','discharged'),
('Simran Kaur','simran@gmail.com','2026-03-09 10:20:00','admitted'),
('Rakesh Kumar','rakesh@gmail.com','2026-03-10 14:35:00','admitted'),
('Meera Iyer','meera@gmail.com','2026-03-11 11:00:00','discharged'),
('Sanjay Patel','sanjay@gmail.com','2026-03-12 16:10:00','admitted');

```

```

INSERT INTO medicines (name, stock, expiry_date) VALUES
('Paracetamol',120,'2027-05-01'),
('Ibuprofen',80,'2026-12-01'),
('Amoxicillin',60,'2026-10-15'),
('Azithromycin',40,'2026-09-01'),
('Metformin',90,'2027-03-01'),
('Aspirin',150,'2027-07-01'),
('Cetirizine',75,'2026-08-01'),
('Dolo 650',110,'2027-02-15'),
('Pantoprazole',50,'2026-11-10'),
('Atorvastatin',65,'2027-01-01'),
('Insulin',30,'2026-06-01'),
('Vitamin D',100,'2027-04-01'),
('Calcium Tablets',85,'2027-08-01'),
('Cough Syrup',55,'2026-09-20'),
('Antacid',70,'2027-01-10');

```

```

INSERT INTO appointments (patient_id, doctor, scheduled_at, status) VALUES
(1,'Dr. Mehta','2026-03-02 10:00:00','completed'),
(2,'Dr. Shah','2026-03-03 11:30:00','completed'),
(3,'Dr. Patel','2026-03-04 09:15:00','scheduled'),
(4,'Dr. Verma','2026-03-04 14:00:00','scheduled'),
(5,'Dr. Gupta','2026-03-05 10:45:00','cancelled'),
(6,'Dr. Mehta','2026-03-05 12:00:00','completed'),
(7,'Dr. Shah','2026-03-06 09:30:00','scheduled'),
(8,'Dr. Patel','2026-03-06 16:00:00','completed'),
(9,'Dr. Verma','2026-03-07 11:00:00','scheduled'),
(10,'Dr. Gupta','2026-03-07 13:45:00','scheduled'),
(11,'Dr. Mehta','2026-03-08 10:30:00','completed'),
(12,'Dr. Shah','2026-03-08 15:00:00','scheduled'),
(13,'Dr. Patel','2026-03-09 09:00:00','scheduled'),
(14,'Dr. Verma','2026-03-09 14:20:00','completed'),

```

```
(15,'Dr. Gupta','2026-03-10 11:15:00','scheduled'),
(1,'Dr. Mehta','2026-03-10 16:30:00','scheduled'),
(3,'Dr. Shah','2026-03-11 10:00:00','scheduled'),
(5,'Dr. Patel','2026-03-11 12:45:00','cancelled'),
(7,'Dr. Verma','2026-03-12 09:50:00','scheduled'),
(9,'Dr. Gupta','2026-03-12 15:10:00','scheduled');
```

```
show tables;
```

```
select * from patients;
select * from medicines;
select * from event_log;
select * from appointments;
```

```
-- one time event:-
-- the hospital is doing a one time data cleanup on a fixed date
-- mark all the appointments older than 1 year as expired.
```

```
delimiter $$
create event expire_old_appointments
on schedule at "2026-03-16 15:50:00"
comment "One time event for data clean up: expire all appointment older than 1
year"
do
begin
    update appointments
    set status = 'expired'
    where scheduled_at < now() - interval 30 day
    and status = 'scheduled';
end $$
delimiter ;
```

```
show create event expire_old_appointments;
```

```
alter event expire_old_appointments
on schedule at "2026-03-16 15:28:00";
```

```
select *, now() - interval 1 year from appointments;
select now() - interval 1 year as "time";
select date_sub(curdate(), interval 1 year);
select timestampdiff(day, "2025-03-16",curdate());
```

```
-- how to see the comments of your events:-
select event_name, event_comment
```

```
from information_schema.EVENTS
WHERE EVENT_NAME = 'expire_old_appointments';
```

```
-- RECURRING EVENT :-
SELECT * FROM EVENT_LOG;
```

```
-- 1  "HEART_BEAT_EVENT_LOG", NOW(), "EVENT SCHEDULAR IS RUNNING FINE"
```

```
CREATE EVENT HEARTBET_LOG
ON schedule EVERY 1 MINUTE
STARTS NOW() + INTERVAL 1 minute
ENDS NOW() + INTERVAL 5 MINUTE
DO
    INSERT INTO EVENT_LOG (EVENT_NAME, RAN_AT, MESSAGE)
    VALUES("HEARTBEAT_EVENT_LOG",NOW(),"EVENT SCHEDULAR IS RUNNING FINE");
```

```
SELECT * FROM PATIENTS;
--
DELIMITER $$
CREATE EVENT ARCHIVE_OLD_PATIENTS
ON schedule EVERY 1 minute
STARTS NOW() + INTERVAL 1 MINUTE
DO
BEGIN
    UPDATE PATIENTS
    SET STATUS = "ARCHIVED"
    WHERE STATUS = "DISCHARGED"
    AND ADMITTED_AT < NOW() - INTERVAL 50 DAY;
END $$
DELIMITER ;
```

```
SHOW EVENTS;
```

```
-- CREATING A ARCHIVE TABLE
CREATE TABLE ARCHIVE_PATIENTS LIKE PATIENTS;
```

```
DESC ARCHIVE_PATIENTS;
```

```
-- ARCHIVE ALL THE PATIENTS WHO HAVE BEEN DISCHARGED FOR MORE THAN 50 DAYS.
DELIMITER $$
CREATE EVENT IF NOT EXISTS ARCHIVE_OLD_PATIENTS
ON SCHEDULE EVERY 1 minute -- 1 DAY
STARTS NOW() + INTERVAL 30 SECOND
COMMENT "USED FOR ARCHIVING OLD PATIENTS"
```

```
DO
BEGIN
    -- ARCHIVING THE PATIENTS INTO ARCHIVE TABLE
    INSERT INTO ARCHIVE_PATIENTS
    SELECT * FROM PATIENTS
    WHERE STATUS = "discharged" AND
    ADMITTED_AT < NOW() - INTERVAL 50 DAY;

    -- deleting patients appointments
    delete from appointments
    where patient_id in (SELECT patient_id FROM PATIENTS
    WHERE STATUS = "discharged" AND
    ADMITTED_AT < NOW() - INTERVAL 50 DAY);

    -- DELETING THOSE PATIENTS FROM PATINET TABLE
    DELETE FROM PATIENTS
    WHERE STATUS = "discharged" AND
    ADMITTED_AT < NOW() - INTERVAL 50 DAY;
END $$
DELIMITER ;
```

```
show events;
select * from appointments;
SELECT * FROM PATIENTS;
SELECT * FROM ARCHIVE_PATIENTS;
```

```
SHOW EVENTS;
```

```
desc appointments;
```